

Generation Interconnection System Impact Study Report

For Queue Project Q22-2066

Virginia Tech Power Plant 138 kV
May 01, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2066. The Interconnection Customer proposes a 215.5 MW natural gas combined cycle facility interconnecting at the Virginia Tech Power Plant 138 kV substation in the Virginia area. The request was received on April 15, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 476 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,209,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HARTMAN CK 138-1 kV TX	HINTON AP – LOWMOOR 345 KV Line	125.4	490	63.7
HARTMAN CRK 345/1 kV Transformer	HINTON AP – LOWMOOR SUBSTATION 345 KV	120.7	509	63.7
MANATEE – HINTON AP 345 KV	HARTMAN CREEK – LONGVIEW2 345 KV	103.9	1,150	29.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HARTMAN CK 138-1 kV TX	Replace transformer with higher-rated unit	\$6,436,000	16% (\$1,047,000)
HARTMAN CRK 345/1 kV Transformer	Replace transformer with higher-rated unit	\$20,408,000	16% (\$3,322,000)
MANATEE – HINTON AP 345 KV	Reconductor line	\$15,709,000	5% (\$840,000)
	Total allocated to project		\$5,209,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.